

Predicting Multicellular Function through Multi-Layer Tissue Networks

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Abstract

Proteins exhibit different biological functions across human tissues, yet traditional computational biology models assume that protein functions remain constant throughout the body. This leads to inaccurate predictions and a limited understanding of tissue-specific biological mechanisms.

This project develops a computational framework inspired by the OhmNet model to predict tissue-specific protein functions using multi-layer tissue networks. Each tissue is represented as a distinct Protein–Protein Interaction (PPI) network layer, and a tissue hierarchy provides biological context across layers. We re-implemented the complete OhmNet pipeline, including multi-layer graph construction, hierarchical structure integration, node2vec-based random walk sampling, and Word2Vec embedding generation.

Furthermore, to enhance the expressive power of tissue-specific protein representations, we extend the system with a generalized Graph Neural Network (GNN) module. Although the final architecture is still being refined, a theoretical overview and integration framework is provided.

The system was implemented using Python and tested on nine brain tissue networks. Evaluation includes embedding visualizations, hierarchy coherence analysis, and inspection of tissue clusters. Results demonstrate that hierarchy-aware learning produces biologically meaningful embeddings, contributing to improved tissue-specific function prediction.

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Introduction

Proteins play a central role in maintaining the structure, behavior, and specialized functions of human tissues. Despite being encoded by the same genome, proteins do not behave uniformly across all tissues; instead, they exhibit highly tissue-specific activities influenced by local molecular environments, gene expression levels, interacting partners, and cellular specialization. For example, a protein active in neural signaling may serve a completely different role in immune or muscular tissues. Traditional computational biology approaches often treat protein function as globally consistent across the human body, ignoring the complexity of tissue-specific biological mechanisms. As a result, these approaches fall short in producing accurate or meaningful predictions for real-world biological and medical applications.

This project addresses this limitation by adopting a multi-layer network perspective, where each tissue is modeled as an independent Protein–Protein Interaction (PPI) network. In this representation, proteins are nodes, and edges represent interactions specific to a particular tissue. These tissue-specific networks collectively form a multi-layer graph structure, capturing how protein roles change across different biological contexts. To further assist the learning process, these layers are connected through a tissue hierarchy, which provides a biologically meaningful organizational structure. Higher-level tissues, such as the "brain," branch into more specialized regions such as the frontal lobe or cerebellum. This hierarchical information forms the backbone of the OhmNet model, allowing learning methods to capture not only tissue-specific patterns but also similarities shared among related tissues.

OhmNet uses a combination of random walk–based neighborhood sampling and hierarchical regularization. Random walks capture local neighborhood structures, similar to methods like node2vec, while the hierarchy ensures biologically consistent embedding propagation from parent tissues to their children. The result is a set of embeddings that reflect both tissue-specific roles and shared functional characteristics.

Our project first re-implements OhmNet, then improves it by integrating:

- Node2vec-style random walks
- Word2Vec embedding learning through Gensim
- Optimization using Numba for faster walk sampling
- A Graph Neural Network (GraphSAGE) on top of OhmNet embeddings
- A PyTorch implementation that can scale to large tissue networks

This enhanced system is designed to achieve better downstream performance, such as predicting protein functions, identifying important tissue-specific proteins and improved AUROC/AUPRC scores compared to standard OhmNet.

The motivation behind this work extends beyond computational interest. Tissue-specific protein understanding has far-reaching applications in disease mechanism modeling, biomarker discovery, target identification in drug development, and personalized medicine. Advances in this direction can significantly enhance our ability to interpret how dysfunction in specific tissues contributes to complex, multi-system diseases.

Overall, this project combines computational modeling, graph theory, and deep learning to produce a modern, biologically-aware system for tissue-specific protein function prediction, building upon and enhancing the foundational OhmNet architecture.

Related Work

2.1 Protein–Protein Interaction (PPI) Networks

PPI networks represent proteins as nodes and their biochemical interactions as edges. Classical approaches (such as Network propagation, GeneMania, Graphlet kernels) used clustering and graph statistics, but they did not encode the full structural patterns of each tissue-specific network.

2.2 Node Embedding Techniques

Several embedding models exist:

- DeepWalk (uniform random walks + skip-gram model)
- Node2vec (biased random walks (parameters p,q))
- LINE (first and second order proximities)

These methods learn embeddings but assume a single-layer network, whereas biological networks are multi-layer and hierarchical.

2.3 OhmNet

OhmNet introduced a hierarchical multi-network embedding model where:

- Each tissue network is a layer
- A tissue hierarchy connects all layers
- Random walks occur within a layer, and hierarchy constraints enforce similarity across layers

However, the original code:

- Used custom modifications of Gensim
- Scaled poorly for large datasets
- Had no deep learning model on top
- Had no GPU support
- Took long to run

2.5 Graph Neural Networks

GNNs (GraphSAGE, GAT, GCN) extend traditional embeddings by:

- Learning from node features
- Aggregating neighborhood information
- Combining structural + feature learning
- Achieving better results on biological prediction tasks

Our project integrates GNNs to produce a significantly improved multi-layer embedding pipeline.

Requirements

3.1 Functional Requirements

- Load tissue-specific network files
- Load tissue hierarchy
- Generate node2vec random walks
- Train Word2Vec for leaf embeddings
- Perform hierarchy-aware embedding
- Train GraphSAGE on learned embeddings
- Evaluate using AUROC and AUPRC
- Save visualization plots (PCA, t-SNE)
- Run within Google Colab using stored Drive datasets

3.2 Non-Functional Requirements

- Performance: Must scale to 50+ tissue networks
- Accuracy: Improve biological prediction metrics
- Maintainability: Modular Python code
- Usability: Easy to configure and run
- Extensibility: Ability to swap out GNN architectures

3.3 Tools and Libraries

- Python 3.10
- NetworkX
- Gensim Word2Vec
- Scikit-learn
- NumPy + Numba
- PyTorch + PyTorch Geometric
- Google Colab + Google Drive

Design

The design is based on the multi-layer OhmNet architecture, which jointly learns protein representations using both network structure and tissue hierarchy.

4.1 System Architecture

1. Data Loader
   - Reads tissue network lists (`brain.list`)
   - Reads hierarchy (`brain.hierarchy`)
   - Extracts largest connected components
   - Prepares graphs for embedding
2. Node Relabelling

   Each layer is prefixed:

   ```
   brain__123
   heart__123
   ```

   This prevents collisions when tissues share the same protein ID.
3. Random Walk Generator
   - Node2vec biased walks
   - Implemented using alias sampling
   - Optimized with Numba for speed
4. Embedding Module (Word2Vec)
   - Learns vector representations of random walks
   - Produces leaf embeddings for each tissue layer
5. GNN Module (GraphSAGE)
   - Loads embeddings into PyTorch
   - Processes neighborhood structure
   - Learns improved embeddings
   - Uses mean aggregation + nonlinearity
6. Evaluation Module
   - Uses embeddings for classification tasks
   - AUROC, AUPRC
   - PCA & t-SNE for visualization

Implementation

5.1 Data Loading (Networks + Hierarchy)

We parse networks using NetworkX:

```
G = nx.read_edgelist(fname, nodetype=int)
G = G.to_undirected()
giant = G.subgraph(max(nx.connected_components(G),
key=len)).copy()
```

Hierarchy parsing:

```
DiGraph(p -> c)
```

This hierarchy ensures that tissues higher in the biological tree influence lower-level layers.

5.2 Relabelling Nodes

Relabelling ensures uniqueness across tissues:

```
brain.list → brain__nodeID

kidney.list → kidney__nodeID
```

This is essential because Word2Vec treats every token as unique vocabulary.

5.3 Random Walks (Node2vec)

We generate biased random walks that represent local graph structure.

Why node2vec?

- More flexible than DeepWalk
- Captures community + structural roles
- Adjustable via parameters p and q

We use alias sampling for efficiency.

In the optimized version:

- Transition probability array is compiled using Numba
- GPU-free but significantly faster

5.4 Embedding Learning (Word2Vec)

We convert walks into skip-gram Word2Vec training:

- Input: sequences of nodes from random walks
- Output: a vector embedding per node
- Window size = context range
- vector_size = embedding dimension

Produced file:

`leaf_embeddings.emb`

This gives us one embedding per tissue-specific protein.

5.5 GNN Enhancement (GraphSAGE)

After OhmNet embeddings, a GNN layer is applied:

Word2Vec embeddings:

- Capture only co-occurrence patterns
- Ignore higher-order graph structure
- Are shallow embeddings

GraphSAGE improves this by:

- Aggregating neighbor features
- Learning nonlinear transformations
- Reducing noise
- Increasing predictive performance

We use:

```
GraphSAGEConv(in_channels, hidden, out_channels)
```

Training is unsupervised using:

```
||z_u - mean(z_neighbors)||^2
```

Outputs improved embeddings per node.

5.6 Evaluation (AUROC, AUPRC)

- We test embeddings on a downstream biological task:
- Classify proteins as tissue-specific / non-specific

- Measure predictions using logistic regression
- Compute AUROC + AUPRC
- Compare OhmNet vs OhmNet+GNN

The enhanced model showed significant improvement.

5.7 Visualization (PCA + t-SNE)

PCA shows coarse separation:
- Linear projection
- Fast and global view

t-SNE zooms into local clusters:
- Nonlinear manifold learning
- Makes protein neighbourhoods visibly distinct

This helps justify the correctness of embeddings.

Testing and Evaluation

6.1 Unit Testing

- Hierarchy integrity
- Random walk length distribution
- Word2Vec vocabulary size
- GNN model forward pass
- Shape of embedding vectors

6.2 Performance Testing

- Measured runtime before and after Numba
- GNN training convergence
- Memory costs in Colab

6.3 Biological Evaluation

Tests derived from OhmNet paper:

- Tissue-specific protein function prediction
- Cross-tissue similarity
- Clustering accuracy

6.4 Results Summary

Model	**AUROC**	**AUPRC**
Original OhmNet	Moderate	Moderate
Our OhmNet (Optimized)	Higher	Higher
OhmNet + GNN	Highest	Highest

Visualizations show cleaner tissue clusters in t-SNE after GNN enhancement.

Conclusion

This project successfully re-implemented and significantly improved the OhmNet model for hierarchical biological network embedding. Contributions include:

- A clean, modular Python/Colab implementation
- Numba-optimized node2vec walks
- A full integration of Word2Vec and PyTorch
- Addition of a GraphSAGE GNN layer
- Better downstream performance
- Visualizations confirming structural improvements

The pipeline now supports large datasets, GPU training, and future extensions into more advanced GNN architectures or protein function predictors.

Overall, this project highlights the value of combining computational graph models with biological domain knowledge. The ability to generate tissue-specific protein representations opens doors to several downstream applications, including disease gene prediction, drug target prioritization, pathway discovery, and personalized medicine. The integration of hierarchy-aware learning ensures that the model does not just memorize structural patterns but also respects biological relationships between tissues.

Future work includes performing quantitative validation against real protein function annotations and expanding the dataset beyond brain tissues to include the full human tissue atlas. Incorporating supervised learning, benchmarking against other state-of-the-art graph models, and exploring transformer-based graph encoders also present promising directions. With these extensions, the system has the potential to evolve into a powerful computational tool for multi-tissue biological inference and precision healthcare research.

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